

FraudGuard AI

Explainable AI for transaction fraud and risk detection

A supervised fraud classifier and an unsupervised anomaly detector score every transaction, combine into one risk score, explain the reasons, and recommend the next action.

Random Forest

Isolation Forest

Risk engine

Streamlit

Team: [TEAM NAME]

Members: [MEMBER 1] · [MEMBER 2] · [MEMBER 3] · [MEMBER 4]

Institution: [INSTITUTION] Event: [EVENT NAME] Date: [DATE]

Every transaction lands in one band

LOW 0-30

Allow

MEDIUM 30-70

Verify

HIGH 70-100

Manual review

Manual fraud review cannot keep up

Four problems a usable fraud system has to solve at once

Volume

Digital payments generate transaction volumes far beyond what a review team can inspect by hand.

Shifting patterns

A classifier trained only on known fraud misses behaviour that has not been labelled yet.

False alerts

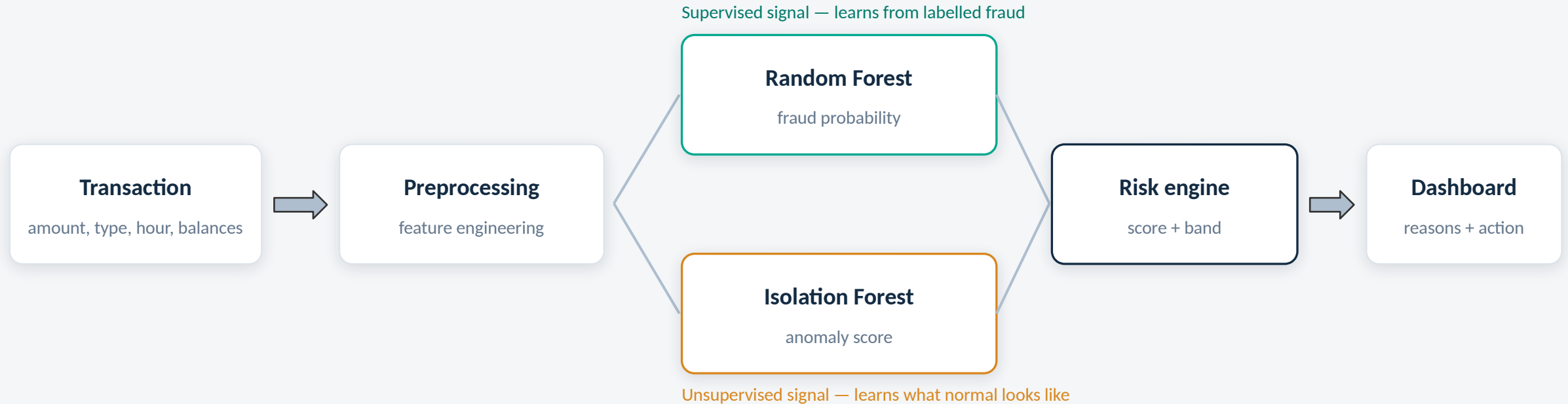
Over-flagging blocks genuine customers and burns analyst time, so precision matters as much as recall.

Unexplained scores

A bare score of 94% gives a reviewer nothing to act on. Decisions need reasons.

One pipeline, two independent signals

Transaction in, risk score with reasons and a recommended action out



Why two models: the classifier catches fraud that resembles past fraud, the anomaly detector catches transactions that simply do not fit the account's normal behaviour. Either one alone leaves a gap.

How the two models work

Both read the same transaction; they answer different questions

1 Random Forest classifier

Question: does this look like fraud we have seen before?

- Trained on labelled transactions from the dataset
- Learns combinations of amount, type and balance behaviour
- Returns a probability-like score via `predict_proba()`

```
fraud_probability = model.predict_proba(txn)[0][1]
```

Read 0.87 as a high estimated likelihood of fraud — not as proof that the transaction is fraudulent.

2 Isolation Forest detector

Question: is this unusual for normal behaviour?

- Needs no fraud labels — models what normal looks like
- Flags transactions that stand apart from the rest
- Raw decision score normalised to a 0-1 anomaly score

```
{"anomaly_score": 0.91, "is_anomaly": true}
```

0 means the behaviour looks ordinary; 1 means it looks highly unusual compared with the rest of the data.

Trained on PaySim, a synthetic mobile-money dataset — a proof of concept, not a production banking engine.

Combining both signals into one risk score

A transparent prototype risk policy — weights and thresholds are our design choices

$$\text{risk_score} = (\text{fraud_probability} \times 0.70 + \text{anomaly_score} \times 0.30) \times 100$$

The classifier carries more weight because it is trained on labelled fraud; the anomaly score adds the behaviour the labels miss.

LOW

0 – 30

Allow, normal monitoring

MEDIUM

30 – 70

Monitor, ask for extra verification

HIGH

70 – 100

Flag for manual review or verification

These thresholds are prototype policy, not industry-calibrated standards. On real operational data they would be tuned to the review capacity and risk appetite of the team using them.

Every score comes with its reasons

The output a reviewer can actually act on

Decision explanation

Fraud probability 91.4%

Anomaly score 94.1%

Combined risk 92.2%

HIGH RISK

Why was this flagged?

- Transaction amount is unusually high
- Transaction pattern is anomalous
- Transaction occurs at an unusual time

Recommendation: MANUAL REVIEW

Illustrative panel from the brief — replace with a live screenshot

Where the reasons come from



Feature values

how far this transaction sits from the account's normal amount, time and balance behaviour



Model feature importance

which inputs the classifier relies on most



Simple business rules

readable checks such as a high amount-to-balance ratio



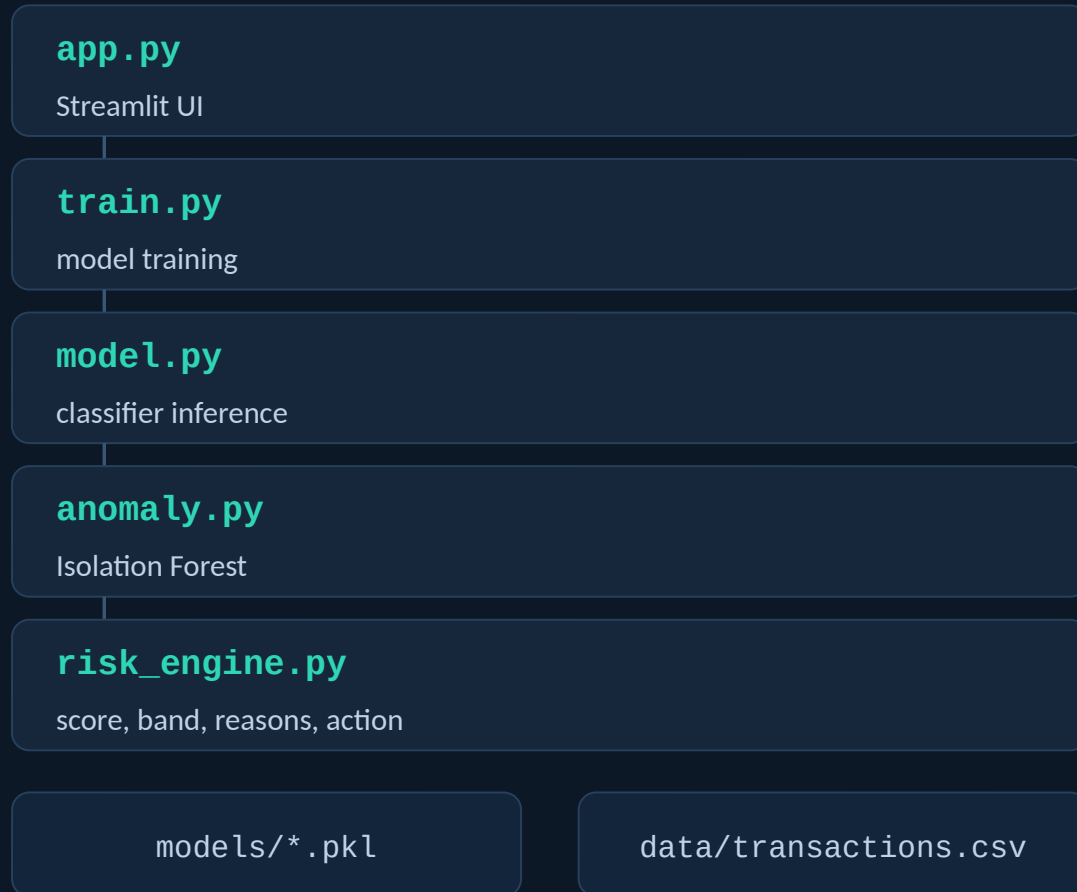
SHAP (optional)

per-transaction attribution, added only if it runs reliably in the time budget

A reason is only shown if the feature behind it actually exists in the data.

Architecture

Five Python modules, one input contract, one output contract



The contract every module agrees on

Input transaction

```
transaction = {
    "amount": 85000, "type": "TRANSFER", "hour": 3,
    "old_balance": 92000, "new_balance": 7000
}
```

Model outputs

```
{"fraud_probability": 0.87}
{"anomaly_score": 0.91, "is_anomaly": true}
```

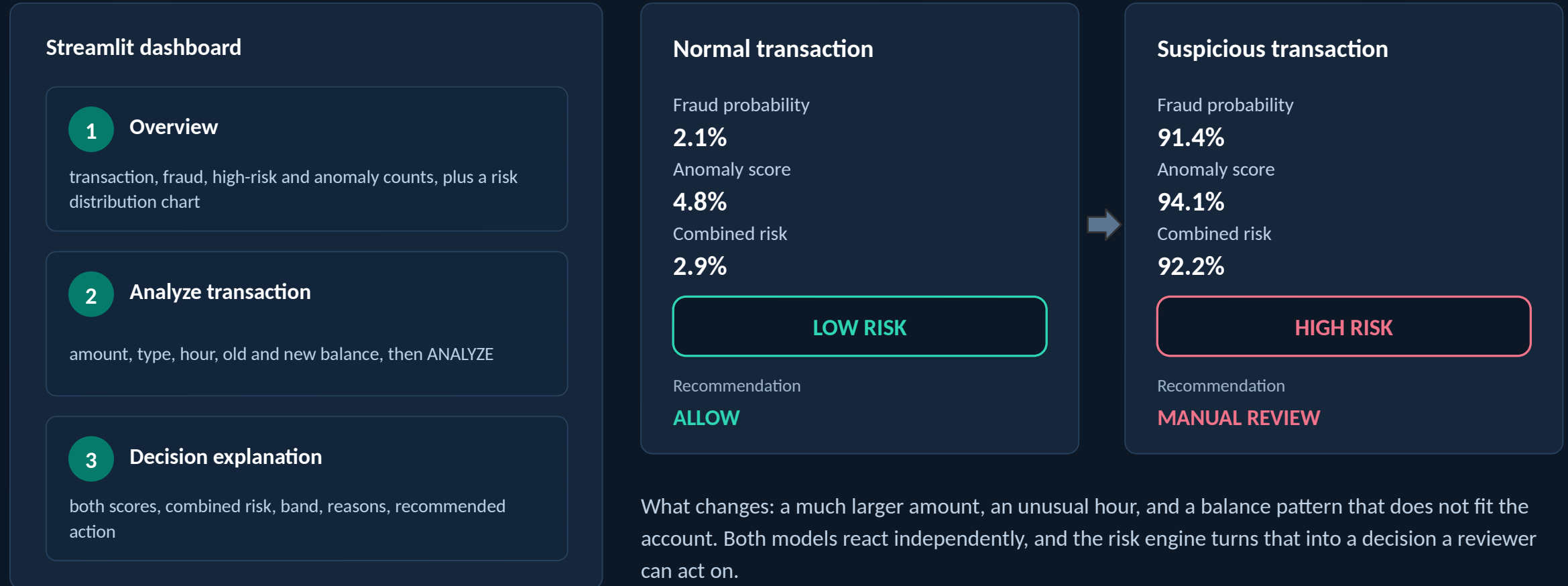
Risk engine output

```
{"risk_score": 88.2, "risk_level": "HIGH",
 "reasons": [...], "recommendation": "MANUAL REVIEW"}
```

Contracts frozen in the first 30 minutes, so four people could build in parallel and integrate without rework.

The product, and the demo in one line

A three-panel dashboard, then the same system on two transactions



Mockup — replace with screenshots of the running app

Results, and what we do not claim

Measured numbers only — anything not produced by a training run stays a placeholder

Precision

[INSERT ACTUAL PRECISION]

Recall

[INSERT ACTUAL RECALL]

F1-score

[INSERT ACTUAL F1]

Confusion matrix

[INSERT TP / FP / FN / TN]

How to read them

- Fraud is rare, so accuracy alone misleads — a model that flags nothing still scores well.
- Recall is the fraud we catch; precision is the analyst time we protect.
- The goal is catching fraud while keeping false alerts manageable.
- Class imbalance and balance-field leakage are checked before training.

Limits we state up front

- PaySim is synthetic; real banking behaviour differs.
- The 70/30 weighting and thresholds need calibration on operational data.
- A model score is evidence, not proof of fraud.
- High risk should trigger verification or review, never an automatic accusation.

Don't just flag it. Explain it.

FraudGuard AI is an explainable transaction-risk system that combines supervised fraud classification with unsupervised anomaly detection to identify suspicious activity, quantify the risk, explain the decision, and recommend the next action.

Two signals

known fraud patterns and unusual behaviour

One score

with a band and a recommended action

Every score explained

reasons drawn from real model inputs

For banks, fintech platforms, payment processors and fraud operations teams — triage by risk band, with the reasons already attached.

Next: real-time scoring, per-account behavioural profiling, graph-based detection, calibration and drift monitoring, human-in-the-loop review.

Thank you — questions welcome.

[TEAM NAME] · [INSTITUTION] · [EVENT NAME] · [CONTACT / REPO LINK]